

HD Pro Design

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HD Car Controller

FOREWORD

- Dear customer, as you know there are car controllers in Unity Asset Store. So why we need to create one more. The requirement to create a controller started with a search to find a controller which includes physical rules and behavior of a real car as in "Forza" or "Need for Speed". We try our best but of course will be happy on hearing from you about new development ideas.
- This car controller has a comprehensive structure. So, before making comments on unity asset store; please read the document carefully for correct settings or write us about any issue which you can't handle.

CONTENT OF THIS DOCUMENT

- You will find a detailed information about HD Car Control what it includes, how it works, what you need to use this controller with its full functionality. In the first section of the document the files in package and the "Base Prefab" of car controller will be defined. In the next section; the files that you need to set initial properties of the controller are listed. In the third section, all properties will be defined and be explained in detail. As last section; works that must be done to be able to run the controller without error.
- The "**Red**" sections in this document mean "**this section is important**". "**Green**" sections mean "**take it easy**".

SHORT INTRODUCTION

- This controller consists of a base rig (will be entitled as "**Base Rig**") which carries all necessary equipment (Body, Wheels Colliders, Lights, Particle Systems etc.) of the car and necessary script/dll files that are responsible to calculate and create all necessary commands.
- A **simple script** attached to the camera in Base Rig. **This script is attached only to handle main camera movements to demonstrate car controller script.** If you want to add comprehensive behavior to the camera please **buy or create your own script**.
- Except one sample motor and environment sound set, sounds used in video introductions are **not included** in to the package.
- **DO NOT FORGET: The 3D design of car is not included in to the package.**

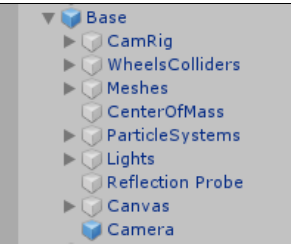
WHAT WILL YOU BUY

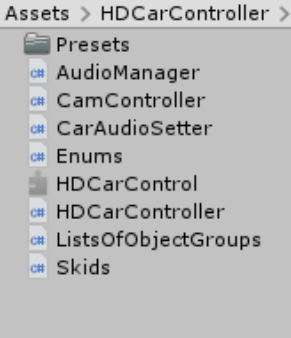
- As you can see from the introduction video, you will buy a script set (**not the cars in the videos**) that controls a mesh group which represent a car.
- With buying this package, you will buy;
 - An engine simulation that includes gears, torque chart, gas pedal efficiency chart, final and rear ratios is very close to real ones.
 - ABS (Anti Blockage System) and TCS (Traction Control System) functions are implemented. **And you may be sure they are very effective.**
 - There is no extra forces on car to simulate a real car. All forces are depends on real physical rules.
 - The resistances are calculated in detail. Air friction, inner friction, wheel friction, down force, terrain specific resistance (such as dirt and soil).
 - A comprehensive sound system which handles engine, gear, tire, shifting and exhaust pat sounds according camera position.

- As you know (or not) a particle system can handle only one material for the particles on air. So what if car changes its terrain while there are particles on air? HD Car Control Particle System will handle with this situation for each tire at the same time for three different terrain. And without any code you can adjust particle materials from car controls.
- There is an “**Output**” class in HDCarController script that you can get nearly all run time data about car.
- And an open source code. **(ABS, TCS and sound fusion is in a DLL.)**

The Base Rig and Script Files of Package

BASE RIG

Base Rig Content	Details
 Figure 1 : Base Rig.	• As shown in the Figure 1 the base rig includes a cam rig that contains the camera and target positions. • WheelColliders game object contains wheel colliders of car. • Meshes section holds 3D Mesh Data of the car. • Center of Mass is the center of car body. • Particle Systems holds exhaust(s) and tire dust particle systems. • Lights section holds lights and canvas section keeps a simple telemetry that gives a general data about car.

Files in the package	Details
 Figure 2: Files.	• Figure 2 shows the list of files. • The main script is HDCarController. Nearly all car properties are controlled by this script. • AudioManager holds the sounds (exhaust pat, turbo, shift up and down etc.) except engine and skid sounds. • CamController is a simple script to manage the camera movements. Do not forget that this script is just for basic camera controls. Please find your own detailed script. • CamAudioSetter is responsible to keep up the camera position index with the HDCarController script. • HDCarControl.dll is a dll file that holds some detailed and intricate math calculations. • Skids carries the code part to create skids on ground. • Other files are Class and Enum holders.

Initial Files (Materials, Sounds etc.)

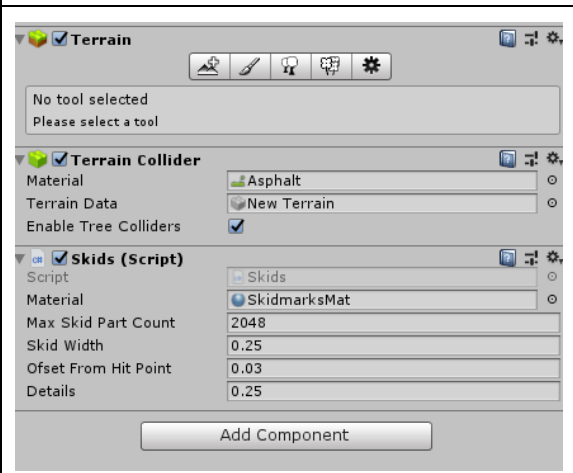
- To set the controller you need some sounds and materials. You have to create physical and other materials of tire settings. In the “Ground Materials” section; to set the frictions of tire dynamically, you have to create physical materials.
- In order to set the dust of tires you need to create dust materials.
- For environmental sounds (turbo sounds, skid sounds, exhaust pat sounds etc.) you have to fill the audio manager with sound files.
- For engine sound you have to use sounds that are recorded different rpms of the car engine.

Steps to Execute the Car

- When you buy this script, most of the properties and values will be adjusted initially.
- In this section all properties will be defined one by one.

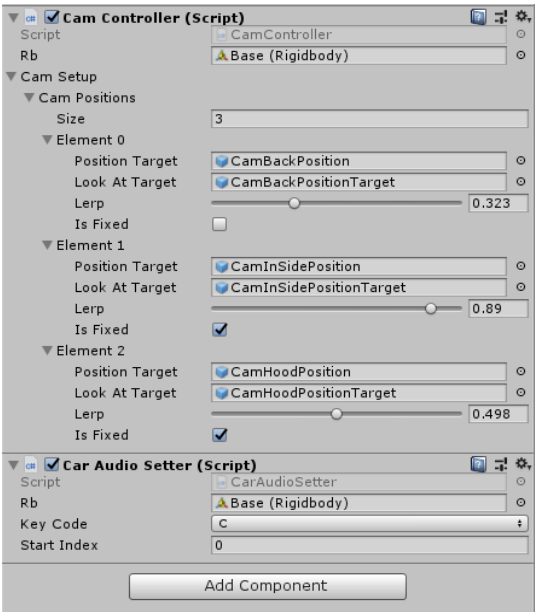
SETTING THE TERRAIN SKID PROPERTIES

- Add the **Skids** script file to the terrain (drag and drop or add component) which you will drive your car on it. **Don't worry about other terrains or objects. Script will automatically handle with this issue.**
- Then select the terrain and under the script add the material for tire skids.
- That is done.

Terrain skid properties	Details
 Figure 3: Terrain Skid Properties.	• Do not forget to define physical material to the terrain(s) which your car will be on it (them). • Material is the material for skids on the ground. • Max Skid Part Count defines the number of maximum triangles for skid trail. • Skid width is width of skids and offset from ground is the distance of the triangles from the ground. • Details is a value of details lower is more detailed.

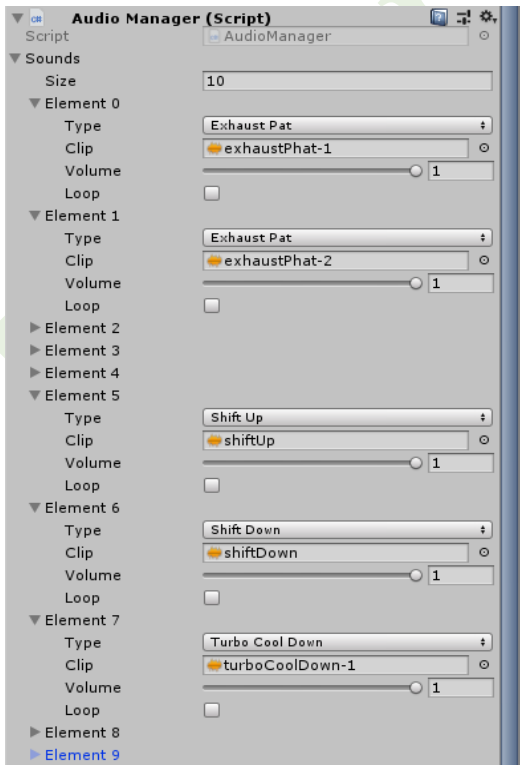
CAMERA CONTROLLER AND CAR AUDIO SETTER

- You can add camera positions and targets as you want. There is no limitation about count and combinations of camera positions and targets.
- HDCarController script has a cam setup section. This section sets the motor and other sound's distortion and low pass filter according to the camera position.
- Let's think like that:
 - You have camera "Positions" and "LookAtTargets".
 - By using your camera script, you will adjust the position and target of your camera.
 - In the HDCarController script there are sound sets in the "Cam Setup" section that adjust sound filters. (Such as for inside, hood and back position.)
 - Camera Audio Setter is responsible to tell the HDCarController script –for example- that "Camera is in second position (For example: inside). So adjust your sound filters according to the second sound adjustment set which is for inside position."
- Simply it matches the camera position with related sound filter set.

Camera Positions	Details
 Figure 4 : Cam Positions and Car Audio Setter.	• In Cam Setup section of Cam Controller script you can define position and target set as much as you want. Just be sure that you defined a sound set in HDCarController to correspond cam position. • In Cam Audio Setter “Rb” defines the rigid body of car. • Key code must be the same with the key code that you use to change the cam position. • Start index defines the initial position index of camera position.

AUDIO MANAGER SCRIPT

- Audio manager script holds the sound clips for car environment (not engine and skid sounds) and properties that belongs to the clip.
- There are four type of sound for car. Exhaust path, turbo cooldown, shift up and shift down.
- You have to define the sound type as necessary. Exhaust pat and turbo cool down sound can be added more than one. The audio system will select sound randomly each time the sound needed.
- **Audio manager initially filled with sample sounds.**
- **These fields are necessary.**

Audio Manager	Details
 Figure 5 : Audio Manager.	• Type defines sound type. This affects the selection of sound for necessary position. If you select exhaust pat type for a sound clip it will be used when an exhaust pat event happens. • Clip is the sound file. • You can adjust the volume level of sound clips individually by using volume property. • Select loop if sound needs to be replayed. (This property is reserved for other environmental sounds (such as motor malfunction, part friction, wind sound etc.) that will be implemented later.)

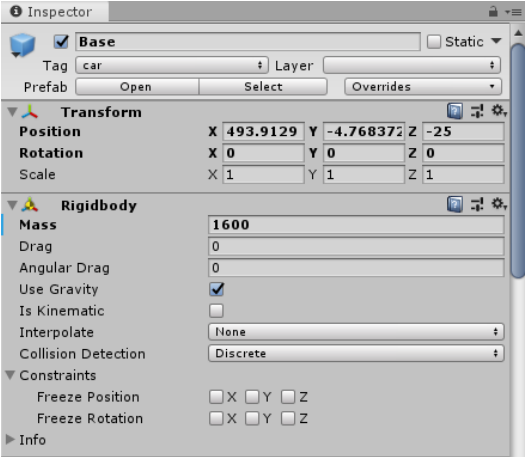
MAIN SCRIPT (HDCARCONTROLLER)	This script will be a backbone for your car. This will make your car alive. Now let's see what's inside.
Main Script	Details
	• Rigid Body Mass is the weight of your car. • Please leave the Drag and Angular Drag as they are.

Figure 6 : Rigid Body Properties.

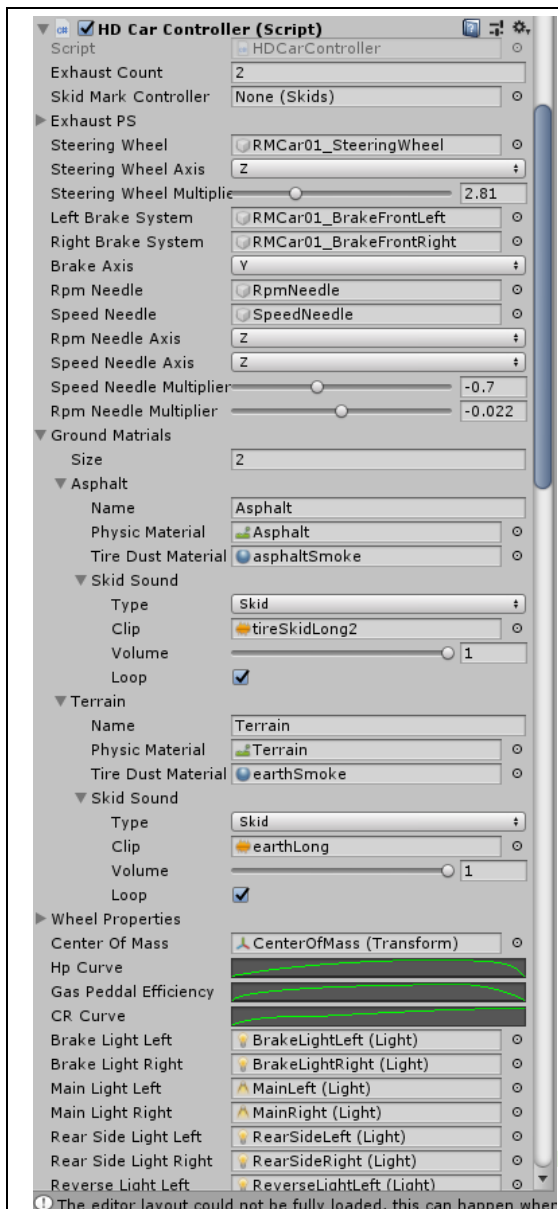


Figure 7 : Exhausts and Ground materials section.

- Exhaust Count can be **one or two**. There are two different particle systems for exhaust pat. **Please do not start particle systems manually. They will adjust themselves according to the sound length added as "exhaust pat" in to the audio manager script.**
- Exhaust PS is the particle system that will be used for exhaust pat.
- Steering wheel is the game object used to simulate steering wheel of car. Steering angle of steering wheel is equal to "Steer Angle of Wheel * Steering Wheel Multiplier".
- Left and Right brake systems are game objects that represents brakes. These parts are the parts need to be turned on "y" axis. (When you turn steering, brakes also have to be turn with wheel. This game object will be turned according to the wheel.)
- The Steering Wheel axis, Brake axis, Rpm Needle axis and Speed Needle axis are the axis that defines on which axis the object will be rotated.
- Speed and Rpm needle multipliers are define that how much the needles are to be rotated from the initial position. Rpm rotation will equal to Engine Rpm * Rpm needle multiplier and speed rotation will equal to max speed * speed multiplier. This means:

$$\text{Rpm Multiplier} = \frac{\text{Angle that the needle has to be rotated}}{\text{Max Engine Rpm}}$$

$$\text{Speed Multiplier} = \frac{\text{Angle that the needle has to be rotated}}{\text{Max Speed Value on Indicator}}$$
- If your car does not have steering wheel, brake parts or needles, just create an empty game object and attach it to the script.**
- Ground Materials section defines on which terrain, which material will be used for dust particle systems, which values will be used for wheel frictions (Physical Material) and also which sound will be played as wheel skid sound. Just define physical materials (adjust dynamic and static friction) and assign them to the terrains. Script will automatically change the skid sound, smoke type and friction values for tires.

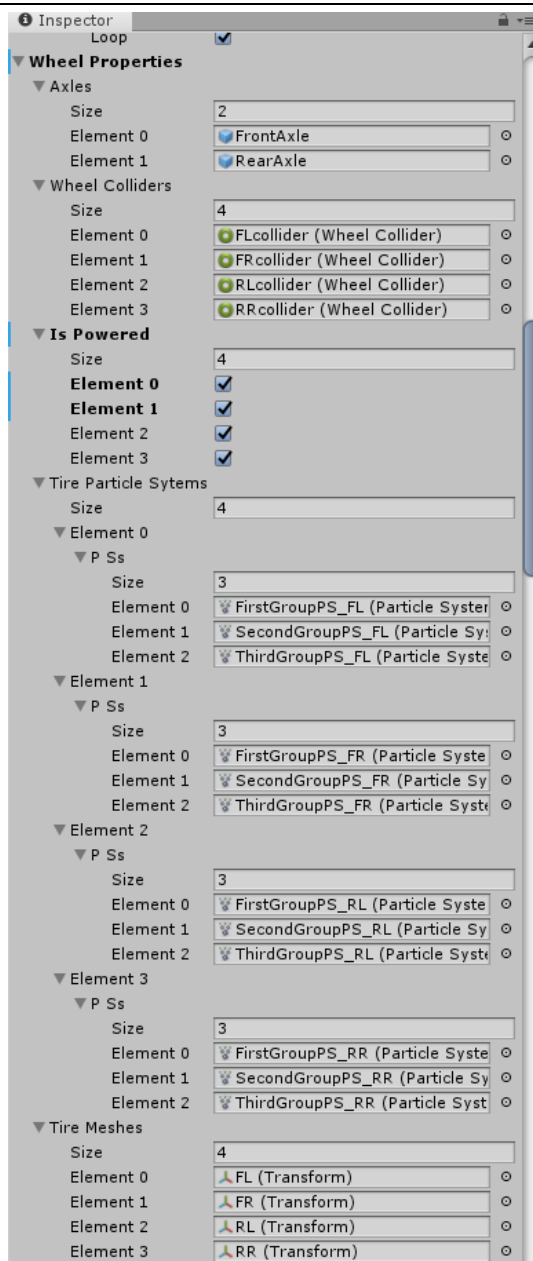


Figure 8 : Wheel Properties.

- Axles have to be **"2"** for this script. This game objects are used to calculate steering angles of front wheels according to Ackerman Rule.
 - For the further fields of "Wheel Properties" the order of wheels must be the same. In initial settings, it is adjusted as Front Left (FL), Front Right (FR), Rear Left (RL) and Rear Right (RR).
 - "Is powered" section defines if the wheel is powered?
 - Tire particle system section may be seen a little bit complex. But it is very simple. In unity, a particle system can only support one material for the particles on air. So, what if tires changes its terrain while other particles still on air? HDCarController script can handle three different tire smoke at the same time. Each element represent a wheel in an order related to the wheel collider order at the top of this section. Each wheel has to have three particle system (First, second and third group for each wheel (FL, FR, RL and RR)).
 - Tire meshes are 3d objects for wheel colliders.
- Don't forget, all orders of this section must be the same order with wheel colliders. For example; if you ordered wheel colliders as FL, FR, RL and RR, particle systems and tire meshes are also have to be at the same order.

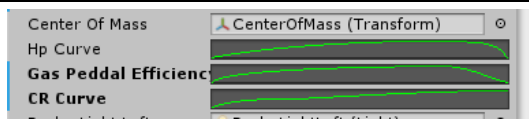


Figure 9 : HP, Pedal Eff. and CR Curves.

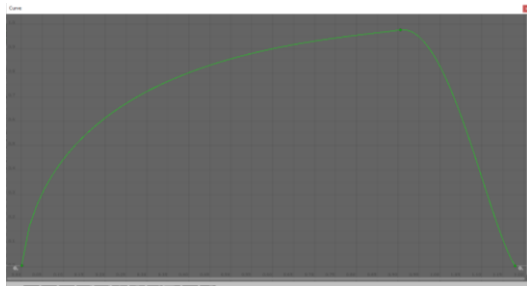


Figure 10 : A sample curve.

$$\text{Torque}_{Nm} = \frac{\text{Power in HP} \times 9,550}{\text{Speed in RPM}}$$

Figure 11 : Torque, Rpm and HP Formula.

- In “HP Curve”, “Gas Pedal Efficiency” and “CR Curve” section you will see a chart on x and y axis.
- **All three curves must be set between 0 and 1.**
- For HP Curve; x axis represents Engine Rpm and y axis represents HP. Chart creates HP values between 0 and 1. Current HP is calculated as $\text{MaxEngineHP} * \text{Evaluated Chart Value}$.
- Example:
 - $\text{MaxEngineHP} = 600 \text{ Hp}$,
 - $\text{Current Engine Rpm} = 4200$
 - $\text{Chart HP Value} = \text{Chart.Evaluate}(\text{Current Engine Rpm}) = \text{Chart.Evaluate}(4200) = 0.8$
 - $\text{Current Produced HP} = 0.8 * 600 = 480 \text{ Hp}$
- As you know there is a throttle value and it depends on engine rpm. On low rpm engine cannot create necessary torque value because of combustion speed. According to the HP-Torque-RPM Conversion formula (Figure 11) in low rpm position, engine should produce so much power.
- Here is an example for an engine which is 600 Hp and runs at 1200 rpm. According to the formula, torque should be 3560 NM. But as you can see in torque charts of cars this is not possible.
- Gas pedal efficiency chart simulates the deficiency at low rpm's.
- CR curve is clutch rate curve. The x axis is the clutch pedal position changes between 0 and 1, y axis is the friction rate of the clutch and changes between 0 and 1. **(The clutch tested with key board but not tested on console that has clutch pedal. This property will be tested in next versions.)**

Brake Light Left	BrakeLightLeft (Light)	○
Brake Light Right	BrakeLightRight (Light)	○
Main Light Left	MainLeft (Light)	○
Main Light Right	MainRight (Light)	○
Rear Side Light Left	RearSideLeft (Light)	○
Rear Side Light Right	RearSideRight (Light)	○
Reverse Light Left	ReverseLightLeft (Light)	○
Reverse Light Right	ReverseLightRight (Light)	○
Signal Light Front Left	TurnSignalLightFrontLeft (Light)	○
Signal Light Front Right	TurnSignalLightFrontRight (Light)	○
Signal Light Rear Left	TurnSignalLightRearLeft (Light)	○
Signal Light Rear Right	TurnSignalLightRearRight (Light)	○
Brake Light Object	SportCar20_BrakeLights	○
Main Light Object	SportCar20_MainLight	○
Rear Side Light Object	SportCar20_RearSideLight	○
Reverse Light Object	SportCar20_ReverseLight	○
Left Signal Light Object	SportCar20_TurnSignalLeft	○
Right Signal Light Object	SportCar20_TurnSignalRight	○
Lights Off Material	MainAlbedo	○
Lights On Material	LightOn	○

Figure 12 : Lights.

- This section defines lights and the meshes that represent the lights.
- You will need two material for light objects. One for lights on position and one for lights off position.

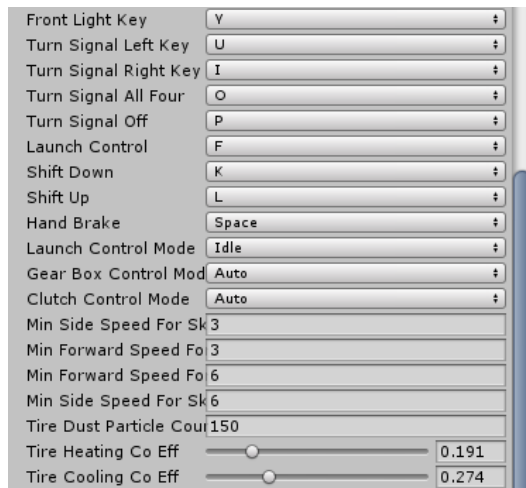


Figure 13 : Key controls and skid values.

- In this section you can set the keys for lights, launch control, shift, hand brake and control modes.
- Car has a Launch Control System. Define your Launch Control Key. First press will make launch control active, and with the second press car will move. Simply press two times to launch control key for a rapid launch.
- Min Side Speed for Skids, Min Side Speed for Skid Volume, Min Forward Speed for Skids and Min Forward Speed for Skid Volume is the variables to decide when skids and skid volumes will be enabled for side and forward wheel skid.
- Tire dust particle count determines the particle intensity for tire dust.
- Tire Heating and Cooling Co Efficiency values determines how fast your tires will heat up and cooldown. Tire temperature will affect the tire dust and skid level.

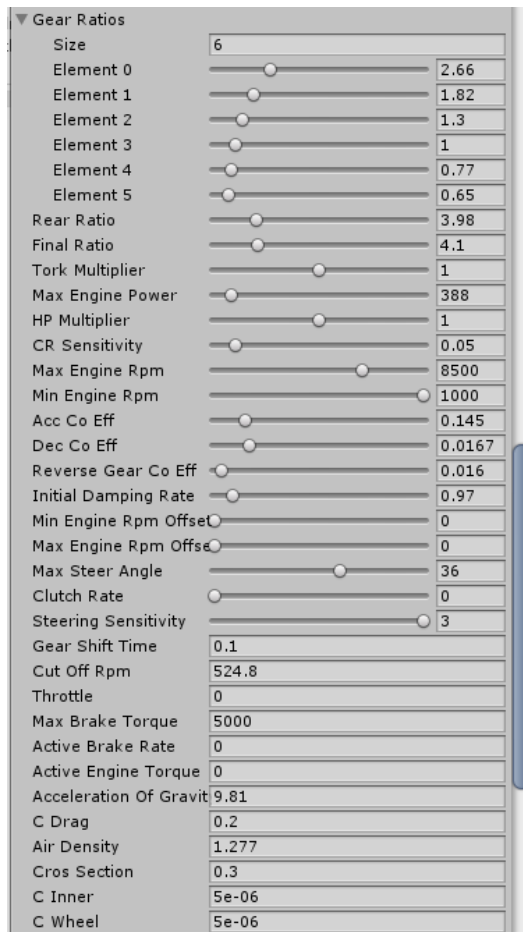


Figure 14 : Engine Properties.

- Gear, Rear and Final ratios works as in a real car.
- Please use real logical values or get real values from real car data sheets.
- Torque Multiplier is multiplies the torque. This provides you an easy way to change torque without changing HP and gas paddle charts.
- Max engine power is the power that engine can reach when Engine Current Rpm/Max Engine Rpm rate equal to the max point of HP chart. Simply; current HP at specific Rpm is calculated by multiplying Max Engine Power with the value evaluated from HP chart at that Rpm Rate: Engine Current Rpm/Max Engine Rpm). You may think that this is complex, but it provides a flexible way of changing Max HP without touching charts.
- **Max and Min Engine Rpm are Read only values. These values loaded from the sound files.**
- Acceleration and deceleration values are the values for increasing engine rpm when throttle bigger than zero and decreasing engine rpm when throttle is equal to zero. It means how fast your rpm will increase and decrease according your throttle. Engine HP is main decider to increase the engine rpm.
- Reverse gear Co Eff determines how much speed your car can reach towards back.
- Max Brake Torque is the max torque that your car's brake can apply.
- **Clutch Rate is read only variable that **shows** your clutch pedal position (Not Clutch friction rate (CR)).**
- **Active Brake Rate and Active Engine Torque are also read only values.**
- Gear Shift Time shows how fast your car can change the gear.
- Cutoff RPM is the rpm that shows how many rpm will decrease when engine reaches to the max rpm.
- Other values (Acceleration of Gravity, Friction Co Eff. (C Drag), Air Density, Cross Section of Car, Inner Resistance (C Inner) and Wheel Friction (C Wheel)) are variables and constants that are used to calculate resistance and down force.

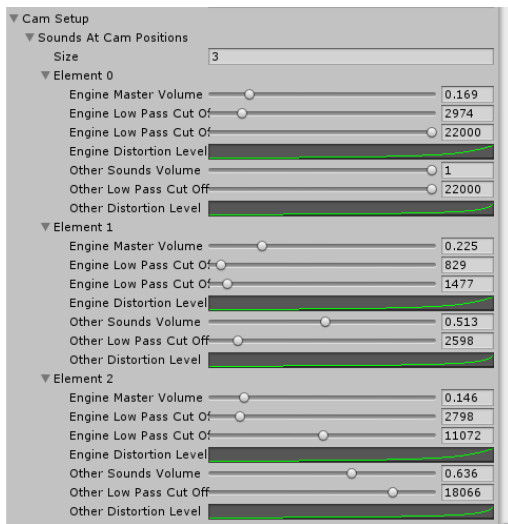


Figure 15 : Sound Setups for camera positions.

- Cam Setup is the section that holds the sound filter parameters for each camera position dedicated in "Cam Audio Setter" script.
- You can adjust engine and other (Tire skid, exhaust pat and turbo cooldown) sound's volume from this section.
- Low pass filter variables adjust the sound's hoarse level. You can envisage it as the difference between the sounds of a car inside and outside. Engine has two low pass filter; one for while engine throttle is bigger than zero (While you are pressing gas pedal.) and one for while throttle is equal to zero (when you release the gas pedal).
- Distortion levels determine the distortion of sounds when they reach their max values. Think, when a car engine reaches its max rpm the sound of engine will distort. You can simulate this with the distortion level parameter. It changes between 0 and 1. The script will map the range to its related sound clip.

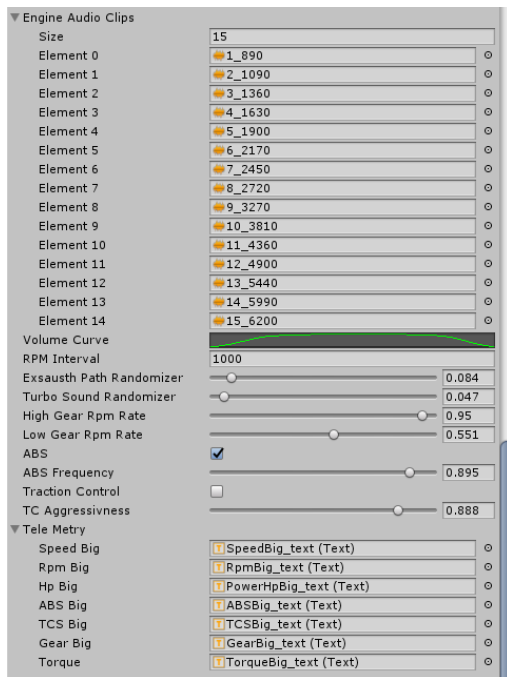


Figure 16 : Engine sound, ABS-TCS and Telemetry Section.

- And the most enjoyable section. HD Car Controller has a comprehensive but simple engine sound controller.
- You must record (or find) engine sound at different rpms with 1000 max interval (bigger intervals are also supported but it will decrease your engine sound quality). Name the files as shown in Figure 16. **Please keep in mind that your max and min rpm values will be determined by the sound file names. So, for the sound set shown in Figure 16 the min engine rpm will be 890 and the max rpm will be 6200.**
- Volume Curve is a curve that determines the mixture type for sound clip parts. Beside, RPM interval determines the range of mixture. **(Most probably these initial values will handle all sets of sounds that you use if you apply the sound record rules mentioned above.)**
- Exhaust Pat and Turbo Sound Randomizers determines how dens the exhaust pat and turbo cooldown sound will be played.
- **High and low gear rpm rates are very important parameters. They determines the gear shift RPM value for shift up and down. For the values shown in Figure 16; the shift down rpm will be $6200 * 0,551 = 3416$ and the shift up rpm will be $6200 * 0,895 = 5890$.**
- ABS is the Anti Blockage System and works as in real cars. It calculates the frictions and wheel skid level and determines the brake level. Frequency determines how fast break torque will be released and applied to the wheels.
- TCS is traction control system and it adjust the torque values to prevent the skid. TCS Aggressiveness is a parameter that determines the amount of intervene of TCS to the torque amounts.
- ABS and TCS are very effective. If you want to drift with the car you should better close the TCS.
- Telemetry section is a simple screen that shows basic car values such as speed and HP. **A detailed output class are defined in the code. You can use it to get the detailed car data.**

Works Must Be Done Before Run the Script

FILES

- To be able to run the code without error, you need materials, physical materials, sound clips (Engine sound clips, exhaust pat, turbo cooldown, skid sounds for each different terrain, shift up and shift down sounds etc.) and meshes (Wheels, steering wheel, light objects and brake system for front wheels).
- **In the package a sample set of these requirements can be found.**

IMPORTANT STEPS

- Please be sure your unity version is updated. The script tested on 2019.2.2f1.
- Be sure you attached skid script to a terrain and defined a skid material.
- Be sure that you attached physical material to the terrain.
- Be sure that you removed all extra scripts attached to the Car Mesh that you downloaded.
- Be sure you removed the Rigid Body that comes with your Car Mesh. As Base Rig has a rigid body you must delete the others.
- Be sure you attached a mesh collider to the body of your car and marked it as convex.
- Be sure all meshes in Wheel Hubs (FL, FR, RL, and RR), Steering Wheel Hub, Rpm Needle Hub and Speed Needle Hub is in zero position, and aligned them as necessary.
- If your car does not have a Game Object which is necessary for HDCarController Script, just create an empty game object and use it.
- Match the cam positions in cam rig with the “Sounds at cam positions” in “Cam Setup” section in HDCarController Script.

Thanks For Buying

We will be glad to hear new ideas or needs that should be implemented in to the car.

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